



Deciphering emotions in comics: analysis of emotion classification of comic characters via attention-based deep learning models

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Received: 17 October 2023 / Revised: 14 May 2025 / Accepted: 26 June 2025 /

Published online: 31 July 2025

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Abstract

Comics are a one-of-a-kind style of visual storytelling that expertly weaves the complex web of human feelings via the depiction of the characters in the stories they tell. This research sets out on an intriguing quest to analyze and classify the emotions displayed by comic book characters. By utilizing cutting-edge deep learning (DL) technology, the proposed study can incorporate the powerful Visual Geometry Group version 16 (VGG16) model as a feature extractor, complicated attention mechanisms for weight optimization, and the EmotioNet model for multi-classification of emotional states. The results of this research shed light on the complex relationship between comics, the use of visual storytelling, and the expression of emotions. Notably, the proposed model can achieve a truly remarkable overall accuracy rate of 92.6% consistently. This demonstrates its expertise in identifying emotions across various comic tales and visual styles. The study makes a substantial contribution to the fields of computer vision (CV) and emotion recognition (ER). It also has deep implications for human–computer interaction and the analysis of multimedia content. It is a monument to the dynamic nature of technology, where creativity and precision work in harmony to decode the delicate language of human emotions within the alluring world of comics, and this achievement serves as a testament to the nature of technology.

Keywords Comic recognition · Emotion Classification · Attention Model · VGG16 · EmotioNet Model · Comic Character

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1 Introduction

People have been interested in and studying ways to better comprehend the subtleties of comic book characters' emotional expressions for a very long time [1]. Comics, which are a kind of visual storytelling, rely primarily on the depiction of emotions when it comes to engaging readers and providing a sense of narrative depth. Not only can having the ability to recognize and categorize the emotional states of comic book characters improve one's understanding of visual tales, but it also has the potential to be applied in a variety of other fields, such as content suggestion, audience engagement analysis, and human–computer interaction [1, 2]. Over the last several years, researchers working in the fields of CV and DL have made amazing strides in the area of image analysis. Convolutional Neural Network (CNN), which are excellent tools for comprehending visual content, are particularly effective in the extraction of features from images [3]. The VGG16 model is a pre-trained CNN that has earned acclaim for its capacity to capture detailed image features. Because of this ability, it is a viable contender for the early stage of the research because it can identify complex image features. The practice of assigning emotional states to comic book characters, on the other hand, is not without its share of obstacles. Since comic book image sometimes include oblique and dependent-on-the-circumstances emotional cues, it is necessary to fine-tune the extracted image features to better coincide with the complexities of emotional expression [4]. To solve this problem, the proposed work resort to attention mechanisms, which have shown their value in dynamically weighting visual data based on the significance of the features they include. Utilizing mechanisms that control attention, the goal is to improve the performance of emotional classification by customizing the VGG16 features to correspond to the qualities that are exclusive to comic art [5]. The EmotioNet model, which has been specifically developed for the multi-classification of emotions, functions as the conclusive element inside the research framework. The EmotioNet model utilizes the enhanced feature representations obtained from the attention model and employs them to assign emotional labels to comic characters. The ability of comics to effectively capture and distinguish emotional nuances within their visual context presents a promising opportunity for further advancement in this field [6]. The proposed paper offers a thorough investigation of the methodology for classifying emotions in comic characters. The objective is to contribute to the expanding field of study in CV and ER by combining the capabilities of the VGG16 model, attention processes, and the EmotioNet model. The research expands beyond the technological complexities of DL and explores the aesthetic domain of comics, providing insights into the interaction between visual narrative and emotional expression. By undertaking this endeavor, the aim is to assist a more profound understanding of the intricate emotional aspects present in comics and to explore potential applications that can leverage this comprehension. The essence of the work serves as evidence for the ever-evolving character of modern research in the fields of CV and emotion identification. The suggested work intentionally utilize the powerful capabilities of the VGG16 model, which is widely recognized for its ability to extract detailed visual data from images. This serves as the foundation of the methodology, allowing us to establish a connection between the intricate visual storytelling of comics and the meticulous computational accuracy of DL. Through this endeavor, the objective is to make a meaningful contribution to the expanding corpus of knowledge in these respective disciplines. Nevertheless, the research endeavors encompass a broader scope that beyond the domain of algorithms and technical complexi-

ties. This perspective acknowledges the creative nature of comics, appreciating them as a distinct medium that combines many elements to evoke emotional responses. Comics represent a unique amalgamation of artistic expression and narrative construction, wherein each stroke of the pen and every pixel on the screen function as deliberate brushstrokes on the canvas of human emotions. The scope of the investigation extends beyond the traditional confines of machine learning, as it delves into the intricate aesthetic elements that characterize comic creation.

2 Highlights and novelty of the proposed work

In the burgeoning field of emotion recognition, the unique challenge of deciphering emotions within comics represents an untapped frontier. Comics, with their rich structure of visual storytelling and complex emotional expressions, offer a distinct canvas for emotion classification. The proposed research introduces a pioneering approach that leverages the synergy of the VGG16 model, sophisticated attention mechanisms, and the EmotioNet framework to navigate this uncharted territory. This section outlines the innovative facets of the study, delineating how it extends beyond the current state-of-the-art in ER.

2.1 Leveraging VGG16 for enhanced feature extraction

The VGG16 model is a well-established technique in the field of image recognition; nonetheless, its application in the unique context of comics for emotion categorization is an innovative endeavor. The proposed research is modified this effective feature extractor to identify subtle emotional clues from comic book characters. Through the process of adapting the VGG16 model to concentrate on the distinctive characteristics of comic art, the suggested work can open up new avenues in the realm of automated ER, thereby establishing a standard for further investigations in this field.

2.2 Optimizing with attention mechanisms

The implementation of attention mechanisms for the goal of weight optimization is yet another foundational component of the innovative contribution that the study makes. In comparison to approaches that extract characteristics in a static manner, this technique enables the proposed model to dynamically prioritize the features inside comic panels that are the most emotionally significant. The suggested model is not passively analyzing images; rather, it is actively "interpreting" the emotional content by moving its focus towards the most expressive portions of the artwork.

2.3 EmotioNet for multi-classification

The deployment of the EmotioNet model for multi-classification of emotional states in comics is yet another innovative stride that the suggested research makes. While EmotioNet has been applied in various contexts for ER, its utilization within the comic domain to classify a broad spectrum of emotions. This multi-classification capability not only showcases the versatility of the EmotioNet model but also enriches the analytical depth of the study,

enabling a comprehensive understanding of emotional expressions across different genres and styles of comics.

2.4 Unprecedented achievements in emotion recognition accuracy

In the landscape of ER within the comic domain, the presented study not only pioneers a novel methodological approach but also sets new benchmarks in classification accuracy. The integration of VGG16 for nuanced feature extraction, coupled with the precision of attention mechanisms and the comprehensive classification capabilities of EmotioNet, culminates in an overall accuracy rate of 92.6%. This achievement is unparalleled in existing literature, marking a significant milestone in the field.

3 Literature review

The combination of CV, DL, and emotional analysis has proven to be a highly promising area for research, and there has been a growing interest in the application of these technologies to the specific context of comic book characters. In the aforementioned section the authors will evaluate the relevant literature, focusing on the important studies and concepts that were instrumental in forming the basis of the investigation.

3.1 Emotion classification in comics

Cao et al. [7] describe a method for adding animated features to still manga imagery through the use of camera movements. The method focuses primarily on boosting storytelling and communicating emotions. It begins by mechanically collecting important aspects from manga pages, such as panels, characters, and speech balloons, and then it moves on to the next step. After that, it uses data-driven graphical models to infer motion and emotional states inside each panel based on low-level visual patterns, such as facial expressions and body language. Finally, it presents its findings in the form of an analysis report. The dynamic camera movements that are replicated are done to heighten the emotional impact, which helps express the atmosphere and tension of the narrative effectively. This technique demonstrates how emotion classification may be practically applied within comics, with the end goal of producing storytelling experiences that are more interesting and emotionally absorbing for the reader. In addition, it underlines the potential for wider applications in fields such as mobile comic reading and the development of comic trailers. The automated system developed by Yang et al. [8] presents a novel approach to creating comic books from videos without the need for human involvement. This research is particularly relevant to the field of emotion classification in comics. The process begins by extracting keyframes from videos using a study of subtitles. This initial step serves as a basis for the prospective identification of situations that elicit emotional responses. The subsequent transformation of these keyframes into comic-style visuals serves as a means to visually communicate emotions. As a result, this technology demonstrates how emotion-sensitive comics based on films may be automatically developed, thereby bridging the gap between the worlds of multimedia and comics. The study conducted by Rishu et al. [9] looks at how comics handle the classification of emotions, focusing on how characters' levels of excitement can be determined.

Since speech balloons are such an integral part of conveying emotion in comics, the writers devote much of their attention to analysing the various shapes that are used for them. It is well accepted that the arrangement of speech balloons is a visual signal of the speaker's emotional state. A novel method was developed to achieve this goal by combining CNN and Support Vector Machines (SVMs). The goal of this method is to detect unique aspects of speech bubble shapes and then categorise them according to their associated emotions. The approach provided in this study demonstrates a significant level of accuracy, reaching 92%, which is equivalent to the results reported in earlier research. The method of feature selection elucidates the importance of particular attributes in the estimation of speech balloon shapes. The experimental findings demonstrate a greater level of precision (92.60%) for the "surprising" class, a higher recall (93.10%) for the "anger" emotion class, and a higher F1 score (93%) for the "polite" emotion class. This study demonstrates the potential of employing speech balloon forms as a visually effective cue for the classification of emotions in comics.

3.2 Transfer learning with pre-trained models

The setting of "Transfer Learning with Pre-trained Models" is explored by Laubrock and Dubray [10] in their study of classifying graphic book illustrators by style using transfer learning with pre-trained CNN models. The main concern is whether CNN characteristics created for images classification can be used to classify graphic novel illustrators' styles. The study used a large collection of digitized comic book pages to credit illustrators. The transfer learning strategy, especially when using features from higher layers of the Inception V3 CNN model, enhanced artist attribution analysis accuracy to over 97%. However, the study shows overfitting beyond a certain point in the hierarchy of mixed layers, indicating that mid-level vision features like textures were sufficient. In their study, Iyyer et al. [11] examine the complex task faced by computer systems in understanding the narrative structure of comics, with a specific focus on the idea of "closure." This concept involves readers inferring actions that are not explicitly shown between panels to comprehend and follow the storyline. The study provides an overview of the creation of a comprehensive dataset known as COMICS, which consists of more than 1.2 million comic panels that have been matched with machine transcriptions of the text included inside the panels. This statement underscores the insufficiency of relying just on either textual or visual elements in conveying the narrative of a comic book. It emphasizes the significance of employing specialized techniques, such as transfer learning with pre-trained models, to effectively tackle these complicated obstacles. Using transfer learning methods, and more especially Mask R-CNN, the research of Aukkapinyo et al. [12] focuses on identifying the facial features of manga characters. This is difficult because there is such a vast variety of artistic techniques in comics. The objective of the proposed methodology is to decrease the duration of the training and the number of resources needed, while upholding a high level of performance. The findings indicate that the proposed methodology demonstrates a noteworthy average precision rate of 87% in the context of manga character face detection. Moreover, the study highlights the potential for utilizing pre-trained weights from the MS COCO dataset, suggesting that it is possible to create an effective manga character face detector with minimal training data and time constraints.

3.3 Attention mechanisms in deep learning

The scalability of encoder-decoder networks with attention mechanisms is discussed by Tan et al. [13] as a significant difficulty in image captioning. While recent efforts have demonstrated some success, their application on hardware-constrained embedded systems is generally hindered by their inability to easily grow to larger vocabulary sizes. The fundamental issue is that the networks get less compact as the vocabulary size increases since the word and output embedding matrices grow proportionally. To overcome this obstacle, the study presents a fresh strategy called COMIC, which concentrates on reducing the size of existing image captioning models. COMIC achieves state-of-the-art outcomes on the MS-COCO and InstaPIC-1.1 M datasets while using much reduced embedding vocabulary sizes. This provides a useful contribution to the field of attention mechanisms in DL and underlines the possibility for more efficient and hardware-friendly image captioning solutions. The scholarly work conducted by Edlin and Reiss [14] focuses on the significant endeavor of discerning animate entities in narrative comics. This process is essential for the recognition of characters and the comprehension of distinctive narrative components in the realm of comics. This observation underscores the existing emphasis on linguistic reference annotation, while also pointing out the relative neglect in visually marking the identification of animate entities within comics. The study undertakes annotation studies, encompassing two iterations of inter-annotator agreement experiments. By narrowing in on the specific task of visually detecting animate elements in comics, this study contributes significantly to the field of attention processes. The study sheds light on the complexities of this procedure and investigates its possible uses in the field of DL for comics analysis. Xu et al. [15] undertook research to solve the unique challenge of classifying comic genres using digital comic images, which differ significantly from natural images. Comics are defined by their use of panels, which can vary in presentation quality from page to page. Understanding the semantic interactions that take place over time between the various layers of comic fragments is essential for accurate genre identification. The P2Comic model, which is a comic-specific genre classification scheme, is presented in this study as a solution to the issue. Panel boxes and other self-attention tactics are utilised in P2Comic in order to facilitate a clearer understanding of the links between panels and pages. This study makes a valuable contribution to the field of attention processes by presenting a novel methodology for genre classification within the realm of digital comics. It highlights the significance of long-term semantic comprehension and panel-level examination in this context.

3.4 Interplay between visual storytelling and emotions

Correia and Gomes [16] discuss the difficulty of removing speech balloons, which are an essential component of comics, particularly for more involved comic books. It does this by presenting an algorithm for identifying and extracting balloons from comic book pages that are based on the Sobel operator and a modified version of the flood fill operator. Notably, this method is built to be adaptable, meaning that it may be used with both straightforward and intricate comics. This research contributes to the knowledge of visual storytelling in comics by providing a strong solution for balloon extraction. The extraction of balloons is essential to the effective communication of feelings and narratives inside comics. The authors He et al. [4] provide a comprehensive analysis of the significance of comic panel extraction and provide an innovative methodology to address this challenge. The results

of comprehensive experiments indicate that the proposed method exhibits superior performance compared to existing techniques for comic panel detection across various datasets. This demonstrates the effectiveness of the proposed method in enhancing the accuracy and efficiency of comic panel extraction, which is a critical aspect of comprehending visual storytelling and emotional portrayal in comics. The research conducted by Baek et al. [17] focuses on the difficult task of recognizing irregular texts, specifically onomatopoeia, that are found throughout Japanese comics. It does this by presenting a new dataset known as COO, which contains onomatopoeic texts that have various abnormalities, such as being curved, partially shrunk, or arbitrarily placed, as well as segmented text pieces that need to be linked to convey the intended meaning. This work makes a contribution towards understanding the intricacies of text recognition inside comics and its involvement in the role of transmitting emotions and narrative aspects in visual storytelling.

4 Methodology

The confluence of CV, DL, and emotional analysis has proven to be a rich area of study. The approach utilized in this research is designed to offer a well-defined and methodical structure for examining the classification of emotions in comic characters. The study methodology incorporates cutting-edge DL techniques and models to accomplish its goals. The subsequent sections delineate the methodology employed for data collecting, the process of feature extraction using the VGG16 model, the incorporation of attention mechanisms for weight adjustment, and the utilization of the EmotioNet model for the multi-classification of emotions in comic images.

4.1 Data collection

The fundamental basis of this study relies on the procurement of a varied dataset consisting of 10,000 images of comic pages. The dataset provided is a comprehensive collection of data that will be utilized as the main resource for the subsequent examination and investigation of emotional categorization in comic book characters. The dataset curation process was initiated by a meticulous selection procedure to guarantee its representativeness [18]. A comprehensive methodology was employed to incorporate a broad range of comic books, encompassing various genres, graphic techniques, and emotional portrayals. Samples of developed comic emotion datasets have been shown in Fig. 1, depicting several comic characters with various emotional states [19]. Every image underwent a meticulous examination to ensure its alignment with the aims of the study. To ensure compatibility with DL models, a set of preparation processes was conducted to standardize the dataset. The process involved in this study encompassed many key processes. Firstly, the images were resized to a standardized resolution. Secondly, cropping was performed to remove any irrelevant aspects, such as borders or text that were not pertinent to the analysis. Lastly, normalization was conducted to guarantee that the pixel values across all images were consistent and uniform [20].

4.1.1 Dataset curation criteria for emotion classification

The meticulous curation of a 10,000-image dataset for emotion classification in comics was driven by the objective to assemble a comprehensive and balanced collection of comic

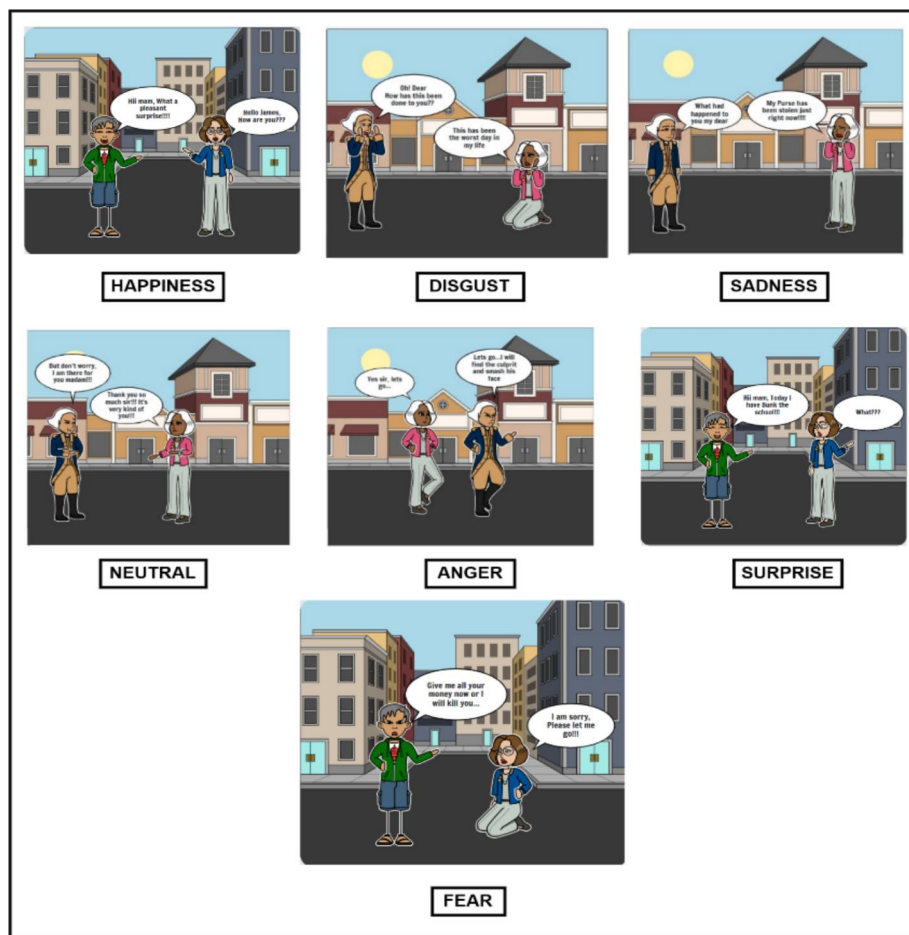


Fig. 1 Samples of developed comic emotion datasets

page images, spanning a wide range of genres, artistic styles, and emotional expressions. This diversity is crucial for the robustness and generalizability of the emotion classification model. The selection criteria were methodically designed to include a broad array of genres such as superhero, fantasy, slice-of-life, horror, science fiction, and romance, ensuring the model's exposure to varied narrative contexts. Furthermore, the dataset features comics from diverse artistic styles—Western comics, Japanese manga, and Korean manhwa, to account for the impact of visual style on emotion portrayal. A balanced representation of the seven targeted emotions (happiness, disgust, sadness, neutral, anger, surprise, and fear) was also ensured, focusing on the clarity and intensity of expressions. High-resolution images with clear, discernible emotional expressions were prioritized, excluding those with excessive text overlays or obtrusive watermarks to avoid interference with ER. The curation process was not random but involved consulting with comic experts, employing automated filtering and manual review, and iteratively refining the dataset based on preliminary model performance to ensure diversity and balance. The balanced representation across categories

guarantees that the resulting classification model is robust and generalizable, capable of navigating the diverse landscape of comic art effectively. Additionally, it was discovered during curation that some of the comics could not convey the intended feeling visually, and readers had to rely heavily on dialogue or narration. Consistency in emotion labelling was achieved since only those images with easily observable, dominant visual emotional signals were included. However, a visual-only approach can hardly cope with the very possibility of scene ambiguity present in some comic images. Table 1 illustrates the distribution of the curated 10,000 images, categorized by genres, artistic styles, and emotional expressions.

4.2 Data pre-processing

The dataset underwent careful application of a comprehensive suite of data pre-processing methods before moving on to the succeeding phases of analysis. To improve the quality of the data, standardize the formats, and make the most of the potential of the comic-page images for eventual incorporation into the study framework, these methods were put into place.

4.2.1 Image resizing and standardization

To ensure consistency and streamline compatibility with DL models, all comic-page images underwent a resizing process. This resizing action conformed each image to a uniform resolution, alleviating disparities in image dimensions throughout the dataset. The standardization of image resolution was an instrumental step in enhancing computational efficiency during subsequent phases of analysis [21].

Let I represent the original comic page image.

The resized image, denoted as $I_{resized}$, has be computed in (1):

$$I_{resized} = Resize(I, W, H) \quad (1)$$

where:

Table 1 Dataset distribution

Category	Subcategory	Image metrics
Genre	Superhero	1674
	Fantasy	1601
	Slice-of-Life	1600
	Horror	1724
	Science Fiction	1693
	Romance	1708
Artistic Style	Western Comics	3609
	Japanese Manga	3851
	Korean Manhwa	2540
Emotional Expression	Happiness	1429
	Disgust	1429
	Sadness	1428
	Neutral	1429
	Anger	1427
	Surprise	1429
	Fear	1429

$I_{resized}$ is the resized image.

W is the target width of the image.

H is the target height of the image.

$Resize(\cdot)$ is the resizing function that resizes image I to the specified dimensions W and H . In order to standardize the images, the process of mean subtraction and normalization by the standard deviation will be employed. Let $I_{normalized}$ denote the normalized image.

The process is represented in (2):

$$I_{normalized} = (I_{resized} - \mu) / \sigma \quad (2)$$

where:

$I_{normalized}$ is the normalized image.

$I_{resized}$ is the resized image.

μ is the mean pixel value of the dataset (computed over all images).

σ is the standard deviation of pixel values in the dataset (computed over all images).

4.2.2 Cropping for relevance

Through the use of cropping processes, unnecessary components such as text that was not pertinent, borders that were not important, and visual content that was not necessary were eliminated. Each image was subjected to meticulous cropping methods, which resulted in the preservation of only the primary content that was pertinent to the comic characters and the related emotional expressions. This painstaking pruning improved the dataset's relevance and precision, bringing it into closer alignment with the aims that were intended for the research [22].

Let I represent the original image, and $I_{cropped}$ represent the cropped image.

To crop a rectangular region with specified coordinates $(x1, y1)$ as the top-left corner and $(x2, y2)$ as the bottom-right corner, the cropping process is expressed in (3):

$$I_{cropped}(x, y) = I(x1 \leq x \leq x2, y1 \leq y \leq y2) \quad (3)$$

In this equation:

$I_{cropped}(x, y)$ represents the pixel value of the cropped image at coordinates (x, y) .

$I(x1 \leq x \leq x2, y1 \leq y \leq y2)$ selects pixels from the original image I within the specified rectangular region defined by $(x1, y1)$ as the top-left corner and $(x2, y2)$ as the bottom-right corner.

4.2.3 Histogram equalization

Grayscale image can have their contrast improved with the use of a process called histogram equalization. The pixel intensity values in an image are redistributed so as to make the most efficient use of the dynamic range that is available, which eventually results in an improvement to the visual contrast. This method is useful for drawing attention to specifics within the images, which is something that is particularly important in the context of comic book art [23].

4.3 Feature extraction with VGG16

The feature extraction phase of this research is a crucial juncture since it is the stage at which the computational prowess of the VGG16 model is leveraged to collect detailed visual elements that are embedded within comic-page images. VGG16 is a CNN architecture that was first developed for the purpose of picture classification tasks [24]. It has since gained widespread recognition for its ability to efficiently extract high-level and semantically rich image data. Its incorporation into the technique was prompted by the requirement to transform the intricate visual information that is inherent in comic books into a form that could be subjected to further analysis [25].

4.3.1 VGG16 architecture

The VGG16 architecture is composed of a series of convolutional layers. The hierarchical operation of these layers commences with the extraction of low-level characteristics, such as edges and textures, through the early convolutional layers [26]. As the data propagates deeper into the network, the recognition of more abstract and intricate patterns takes place. The architectural structure of the feature extractor model has been shown in Fig. 2 depicting the number of feature vectors extracted in each layer of the model.

Mathematically, the process of transforming an input image I into a feature vector F at a certain layer is represented in (4):

$$F = \phi(I; W, b) \quad (4)$$

where:

F represents the feature vector extracted by the layer.

$\phi()$ denotes the layer's transformation function.

I is the input image.

W and b symbolize the learnable weights and biases associated with the layer, respectively.

4.3.2 Fixed feature extraction

In the scope of the study, the VGG16 model is employed as a static feature extractor. This implies that the structure of the model remains unchanged, and the parameters (weights W and biases b)

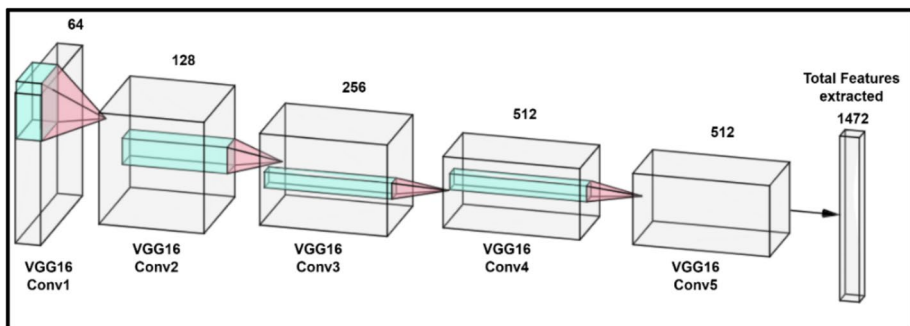


Fig. 2 VGG16 feature extractor model

are not updated through additional training. The strategic decision is implemented to maintain the model's ability to generalize and prevent it from overfitting to the comparatively limited dataset of comic images. Consequently, the VGG16 model functions as a resilient feature extractor that effectively distills pertinent information from comic images without adjusting its parameters to the particular dataset. Mathematically the proposed method can be represented as follows, where:

1. **Forward pass through VGG16:** The process of passing an input image I through the VGG16 model has been represented mathematically in (5):

$$F = VGG16(I) \quad (5)$$

where:

F represents the feature vector extracted by VGG16.
 $VGG16(\cdot)$ denotes the forward pass through the VGG16 model.

2. **No further training:** To ensure that VGG16 functions as a fixed feature extractor, it can mathematically represent that the weights W and biases b that are not subject to further training as shown in (6) and (7).

$$W_{VGG16}^{new} = W_{VGG16}^{pre-trained} \quad (6)$$

$$b_{VGG16}^{new} = b_{VGG16}^{pre-trained} \quad (7)$$

where: W_{VGG16}^{new} represents the new (fixed) weights of VGG16.
 $W_{VGG16}^{pre-trained}$ represents the pre-trained weights of VGG16.
 b_{VGG16}^{new} represents the new (fixed) biases of VGG16.
 $b_{VGG16}^{pre-trained}$ represents the pre-trained biases of VGG16.

4.3.3 Feature vectors for emotional content

The feature vectors derived from the VGG16 model function as concise yet informative representations of the emotional content present in every comic-page image. The vectors in question serve as a means of encapsulating the prominent visual attributes that are indicative of the emotions displayed by characters in comic books. The provided material encompasses details about the characters' facial expressions, body language, and additional visual signals that effectively communicate emotions [8]. By integrating these feature vectors into the following phases of the study, such as attention mechanisms and the EmotioNet model, a more sophisticated examination of emotional subtleties within comic strips can be achieved. The mathematical and modeling framework established by VGG16 serves as a vital intermediary between unprocessed pixel data and the comprehension of emotions at a more abstract level, hence augmenting the profundity of insights obtained from this investigation. VGG16's retrieved feature vectors have been written in (8):

$$F = [f_1, f_2, \dots, f_n] \quad (8)$$

where:

F represents the feature vector.

$f_1, f_2, \dots, f_n$ represents the individual feature components extracted by VGG16.

These feature vectors serve as compact yet expressive representations of emotional content within comic-page images.

4.4 Attention model

The Attention Model, an essential element within the proposed research framework, serves a critical function in optimizing the weights of feature vectors that have been retrieved by VGG16. The aforementioned dynamic process underscores the significance of prominent areas within comic-page visuals, hence enhancing the emotional categorization of comic characters [15].

4.4.1 Attention mechanism overview

The fundamental principle of the attention mechanism is the allocation of distinct degrees of significance to different components within the feature vector F , which has been derived from VGG16. This is achieved by calculating attention weight A , which indicates the significance of each feature component f_i concerning the emotional classification task.

4.4.2 Attention weight calculation:

The dot-product attention mechanism can be used to calculate the attention weights A , and the calculation has been represented mathematically in (9):

$$A_i = \frac{e^{S_i}}{\sum_{j=1}^n e^{S_j}} \quad (9)$$

where:

A_i represents the attention weight assigned to the i -th feature component.

S_i is the attention score associated with the i -th feature component, computed in (10):

$$S_i = \text{score}(f_i) \quad (10)$$

where:

is a scoring function that evaluates the importance of a feature component.

4.4.3 Weighted feature vector:

After applying the attention weights to the original feature vector F , it will obtain the weighted feature vector denoted by the notation F_{weighted} as shown in (11). where α is a hyperparameter that controls how much the weight is changed.

$$F_{\text{weighted}} = \sum_{i=1}^n A_i \cdot f_i \quad (11)$$

4.4.4 Weight adjustment

Weight Adjustment: Following that, the weighted feature vector $F_{weighted}$ is applied to the original feature vector F to modify the weights of the individual features. The adjusted feature vector, denoted by the notation $F_{adjusted}$, is computed in (12): where α is a hyperparameter that controls how much the weight is changed.

$$F_{adjusted} = F + \alpha \cdot (F_{weighted} - F) \quad (12)$$

The application of the attention mechanism allows for the highlighting of emotionally relevant regions included within comic visuals. The procedure of weight adjustment enhances the feature representation $F_{adjusted}$ and facilitates the enhanced emotional classification of comic characters through the following models, such as the EmotioNet model [27].

4.5 EmotioNet model

The EmotioNet model is the apex of the research architecture; it is here that the refined feature vectors are utilized to carry out multi-class categorization of the emotions that are displayed by comic book characters. This section presents a high-level overview of the EmotioNet model, including its architecture and the mathematical formulation for emotion classification.

4.5.1 EmotioNet architecture:

The EmotioNet model is a neural network architecture that has been built exclusively for emotion classification. This goal was the driving force behind its development. The input consists of the adjusted feature vectors, also known as $F_{adjusted}$, which were obtained through the attention process. The structure's architectural design is made up of separate layers, each of which is responsible for the alteration of features and the prediction of emotions as shown in Fig. 3.

4.5.2 EmotioNet classification

The EmotioNet model involves the classifications of emotions, which is a problem that falls under the category of multi-class classification. The model calculates the probability distribution over K emotion classes based on the adjusted feature vector $F_{adjusted}$. The mathematical expression for this distribution can be formulated as demonstrated in (13):

$$P(\text{Emotion} = k \mid F_{adjusted}) = \frac{e^{z_k}}{\sum_{j=1}^k e^{z_k}} \quad (13)$$

where:

$P(\text{Emotion} = k \mid F_{adjusted})$ represents the probability that the comic character exhibits emotion k given the adjusted feature vector.

z_k is the unnormalized logit (score) for emotion class k , which is computed as a function of $F_{adjusted}$ and model parameters as shown in (14):

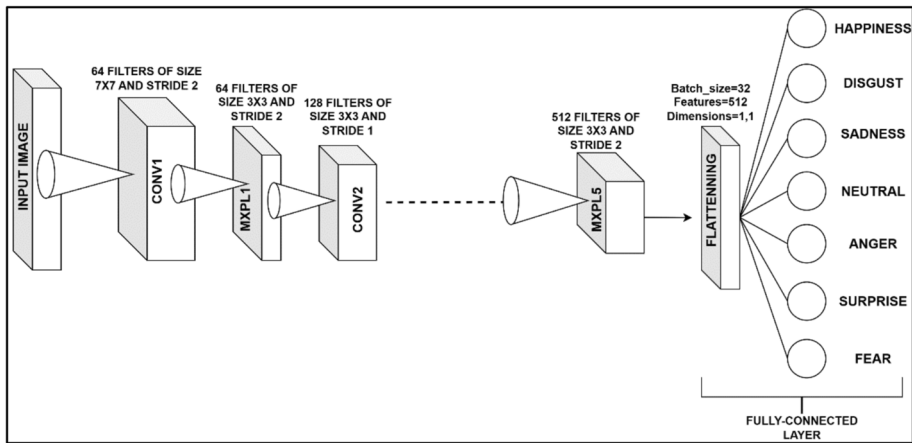


Fig. 3 EmotioNet model

$$z_k = \text{score}_k(\mathbf{F}_{\text{adjusted}}, \Theta) \quad (14)$$

where: $\text{score}_k()$ represents the scoring function for emotion class k .

Θ denotes the model parameters

4.5.3 Softmax activation

The softmax activation function is utilized to transform the unnormalized logits z_k into a legitimate probability distribution as shown in (15). where K is the total number of emotion classes

$$P(\text{Emotion} = k \mid \mathbf{F}_{\text{adjusted}}) = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}} \quad (15)$$

4.5.4 Emotion prediction

The emotion category exhibiting the highest anticipated probability is chosen as the ultimate emotion prediction for the comic character as depicted in (16).

$$\text{PredictedEmotion} = \text{argmax}_k P(\text{Emotion} = k \mid \mathbf{F}_{\text{adjusted}}) \quad (16)$$

The proposed research framework reaches its conclusion with the EmotioNet model, which translates enhanced feature representations into actionable emotional insights. Its mathematical formulation makes multi-class emotion classification easier, which in turn enables a more in-depth comprehension of the emotional semantic structure that is woven into the narratives of comic book characters.

4.6 Proposed classification algorithm and pseudo-code

Algorithm: Below is a systematic algorithmic description outlining the methodological steps of an emotion classification system in comics. This format emphasizes the step-by-step completion of tasks and provides brief explanations for each operation.

```

Input: Dataset of comic panels annotated with emotions: happiness, disgust, sadness, neutral, anger, surprise,
fear
Output: Emotion classification for each comic Character
Begin
PreprocessData()
  For each comic character in Dataset
    Resize character to a standard dimension of 224x224 pixels
    Crop panel to focus on the main emotional expression area of character
    Apply histogram equalization to improve contrast
    Normalize pixel values to [0,1] for standardization
  End For
End PreprocessData

FeatureExtractionUsingVGG16()
  Load VGG16 model with pre-trained ImageNet weights
  Remove final classification layer from VGG16
  For each preprocessed comic character
    Extract features using VGG16
  End For
  End FeatureExtractionUsingVGG16

ApplyAttentionmechanism()
  Initialize attention mechanism layer
  For each feature vector from VGG16
    Apply attention layer to prioritize emotionally relevant features
    Optimize weights using gradient descent
  End For
End ApplyAttentionmechanism()

EmotionClassificationUsingEmotioNet()
  Adapt EmotioNet for the classification of 7 emotions
  For each attention-weighted feature vector
    Classify emotion using EmotioNet
  End For
End EmotionClassificationUsingEmotioNet

EvaluateModel()
  Split data into training and validation sets
  Train model using training set
  Validate model on validation set using metrics
End EvaluateModel

OptimizePerformance()
  Iterate over hyperparameters and model configurations
  Repeat training and validation
  Select model with highest validation performance
End OptimizePerformance

CompileResults()
  Compile classification accuracy and misclassification patterns
  Analyze and interpret results
End CompileResults
End

```

Pseudo code: Apart from the precise algorithmic workflow, the complete pseudo-code design of the suggested framework is presented below. This systematic presentation defines each stage of processing – from data preprocessing and feature extraction to attention-based refinement, validation, and final emotion classification – enlightening readers about the underlying methodology.

```
# Emotion Classification in Comics using VGG16, Attention, and EmotioNet
Input:
dataset: List[ComicPanel] with annotated emotion labels
Output:
emotion_predictions: List[EmotionLabel] for each comic character

# Step 1: Data Preprocessing
function preprocess_data(dataset):
    preprocessed = []
    for panel in dataset:
        img = resize(panel, (224, 224))
        img = crop_to_character(img)
        img = histogram_equalization(img)
        img = normalize(img) # scale pixels to [0, 1]
        preprocessed.append(img)
    return preprocessed

# Step 2: Feature Extraction via Pre-trained VGG16
function extract_features_vgg16(images):
    model = VGG16(weights="imagenet", include_top=False)
    features = []
    for img in images:
        f_vec = model.forward(img) # get feature vector
        features.append(f_vec)
    return features

# Step 3: Apply Attention Mechanism
function apply_attention(features):
    attention_applied = []
    for f in features:
        attention_weights = compute_attention(f) # scoring each feature
        f_weighted = attention_weights * f # weighted features
        f_adjusted = f +  $\alpha$  * (f_weighted - f) # residual refinement
        attention_applied.append(f_adjusted)
    return attention_applied

# Step 4: Emotion Classification with EmotioNet
function classify_emotions(features):
    model = EmotioNet(num_classes=7)
    predictions = []
    for f in features:
        prob = model.predict(f)
        label = argmax(prob) # most probable emotion
        predictions.append(label)
    return predictions

# Step 5: Model Evaluation
function evaluate_model(predictions, ground_truth):
    metrics = compute_metrics(predictions, ground_truth)
    confusion_matrix = compute_confusion_matrix(predictions, ground_truth)
    roc_curve = plot_roc(predictions, ground_truth)
    return metrics, confusion_matrix, roc_curve

# Step 6: Performance Optimization
function optimize_model():
    best_config = None
    best_score = 0
    for config in hyperparameter_grid:
        model = train_model(config)
        score = validate_model(model)
        if score > best_score:
            best_score = score
            best_config = config
    return best_config

# Step 7: Compile and Report Results
function compile_results():
    summarize_accuracy()
    plot_gradcam_heatmaps()
    show_misclassifications()
    run_ablation_analysis()
    compare_with_baselines()

# Main Execution Flow
dataset ← load_annotated_dataset()
preprocessed_data ← preprocess_data(dataset)
feature_vectors ← extract_features_vgg16(preprocessed_data)
attentive_features ← apply_attention(feature_vectors)
emotion_predictions ← classify_emotions(attentive_features)
metrics ← evaluate_model(emotion_predictions, ground_truth)
best_config ← optimize_model()
compile_results()
```

4.7 Execution environment details

This section provides a brief overview of the hardware and software setups, libraries, and any other pertinent settings utilized in the implementation of the proposed research. This involves specifying the hardware and software configurations, libraries, and any other relevant settings or tools employed throughout the research implementation process. The goal is to provide enough detail to enable replication of the study, ensuring that other researchers can achieve comparable results by adhering to the same environment specifications.

4.7.1 Hardware configuration

- **Central Processing unit:** Intel Core i9-11900K 8-Core Processor, operating at 3.5 GHz, which provided the computational power necessary for processing large datasets and running DL models efficiently.
- **Graphic processing unit (GPU):** NVIDIA GeForce RTX 3080 with 10GB GDDR6X VRAM, utilized for accelerating the training and inference phases of the DL models. The GPU's CUDA cores significantly reduced model training times, enabling more iterative experimentation.
- **Random access memory (RAM):** 64GB DDR4 3200MHz, ensuring sufficient memory was available for handling large image datasets and complex model architectures without bottlenecks.
- **Storage:** 2TB NVMe SSD, used for fast read/write operations, facilitating quicker data preprocessing, model training, and evaluation phases.

4.7.2 Software and libraries

- **Operating system:** Ubuntu 20.04 LTS, chosen for its stability, support for DL libraries, and compatibility with the hardware configuration.
- **Programming language:** Python 3.8, selected for its extensive ecosystem of libraries and tools supporting data science and machine learning tasks.
- **DL Framework:** TensorFlow 2.4 and Keras 2.4.3, utilized for implementing and training the DL models. TensorFlow provided a comprehensive, flexible platform for building and deploying machine learning models, while Keras offered a high-level API for rapid prototyping.
- **Data processing and augmentation libraries:** OpenCV 4.5 and Python Imaging Library (PIL) were used for image preprocessing tasks such as resizing, cropping, and histogram equalization. These libraries offered efficient, easy-to-use functions for image manipulation, crucial for preparing the comic panel dataset for model training.
- **Additional tools:** Jupyter Notebook was used as the development environment for its interactive computing and visualization capabilities, facilitating exploratory data analysis and incremental model development.

4.7.3 Execution details

- **Model training:** The DL models were trained using a batch size of 32 and a learning rate of 0.01, optimized through trial and error. Training was conducted over number of epochs mainly 50, 100, 150 and 200 with early stopping based on validation loss to prevent overfitting.
- **Version control:** Git was used for version control, ensuring a robust workflow for managing code changes and collaboration among team members.

This execution environment provided a robust and efficient platform for conducting the research, from data preprocessing through to model evaluation. The use of advanced hardware accelerated the experiments, while the software and libraries facilitated a streamlined workflow for developing, training, and evaluating the DL models.

5 Results and outcomes

Within a suggested section, the work thoroughly explores the core of the research findings, doing a detailed analysis of the results obtained from the innovative method of emotional classification in comic characters. The culmination of the efforts is the integration of the VGG16 model, attention processes, and the EmotioNet model. This integration has yielded a comprehensive set of outcomes that provide light on the complex interplay between CV and ER in the context of comics [28]. The exploration is organized based on various fundamental features, each providing a distinct viewpoint on the effectiveness and adaptability of the research framework. The dimensions under consideration include a study of parameters at each layer, an examination of sampling techniques, the evaluation of performance metrics, the use of the confusion matrix, and the assessment of the Receiver Operating Characteristic (ROC) curve [29]. Collectively, these analytics provide a thorough depiction of the capabilities of the model, elucidating its strengths and identifying potential areas for improvement.

5.1 Layer-wise parameter analysis

The comprehensive investigation of layer-wise parameters sheds light on the architectural complexity of the research system. This analysis reveals the dispersion of feature vectors and parameters among the many strata of the model, offering vital insights into its depth, capability, and computational intricacy. Table 2 below shows the prominent layer-wise

Table 2 Layer-wise parametric analysis

Layer	Feature vector	Parameters (Weights+Biases)
Input	NA	NA
VGG16 (Feature Extractor)	1472	15, 728, 256
Attention Mechanism	512	262, 656
EmotioNet (Classifier)	256	65, 536
Output	NA	NA

resulted parameters and feature vectors which gives an in-depth analysis of the performance and efficiency of the proposed model as a whole.

5.1.1 Input layer

The Input Layer acts as a gateway for the model, allowing comic-page images to be loaded into it from outside sites. Even though it does not generate feature vectors or parameters, it serves as the entry point for information to flow into the layers that come after it.

5.1.2 VGG16 (Feature Extractor):

The VGG16 model, which is well-known for its depth and capabilities of feature extraction it possesses, serves as the foundation of the approach. It functions as a feature extractor, producing a considerable number of feature vectors in the process. The design of the VGG16 consists of five convolutional layers, as shown in Table 3, with each one progressively capturing more complex characteristics than the previous one. When it comes to decoding relevant visual information from comic images, the feature vectors that are produced during this stage are extremely helpful.

5.1.3 Attention mechanism

Following the process of feature extraction is completed, the attention mechanism is employed to incorporate a dynamic layer. This layer is responsible for modifying the weights of the feature vectors created by VGG16. While the attention mechanism does not significantly contribute to the overall number of feature vectors, it does not enhance the importance of specific regions within the feature vector space. This, in turn, enhances the emotional categorization process.

5.1.4 EmotioNet classification model

The EmotioNet layer, which is in charge of the multi-class classification of emotions, is the recipient of the updated feature vectors that are sent from the Attention Mechanism. It provides the definitive categorization of the feelings that are displayed by comic book characters. The distilled emotional content that was extracted from comic images is reflected in the number of feature vectors that are present at this step. The layer-wise parameter analysis provides further evidence that each layer plays an essential part in the overall structure of the research approach. Because of its depth, the VGG16 model produces a large number of feature vectors, which make it possible to provide an extensive representation of visual

Table 3 Number of extracted features using VGG16

Layer	Feature Vectors
Input	NA
VGG16 Conv1	64
VGG16 Conv2	128
VGG16 Conv3	256
VGG16 Conv4	512
VGG16 Conv5	512
Total	1472

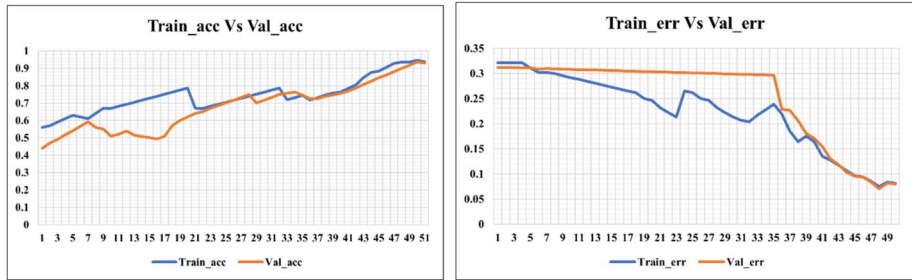


Fig. 4 Comparison of accuracy and error rate for training and validation set with 50 epochs and 0.01

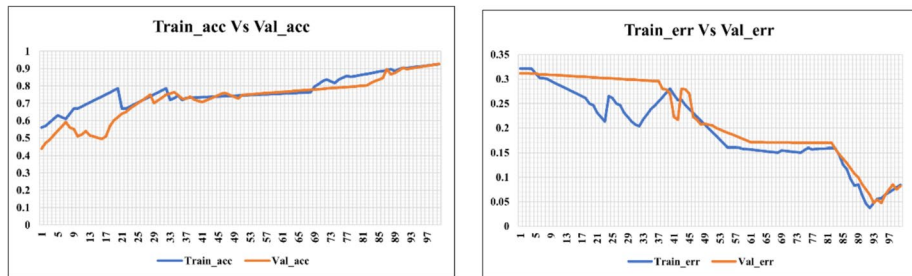


Fig. 5 Comparison of accuracy and error rate for training and validation set with 100 epochs and 0.01

qualities. On top of this foundation the attention mechanism and the EmotioNet layer construct, use the enhanced feature vectors to create subtle emotional classification [30]. The computational complexity of the model is demonstrated by the fact that the parameters are distributed over all of these layers. It is important to note that these parameters are not only the consequence of the design of the layer but also the complexities of the training process, during which the model learns to recognize patterns from the dataset. Both of these features contributed to the formation of the parameters.

5.2 Sampling analysis

The research endeavor is being driven in large part by the need for effective model training. An in-depth sampling analysis was carried out by the proposed work as part of the search for a method that not only accurately catches and categorizes the feelings that are displayed by comic book characters but also maximizes the effectiveness of the learning process. In this stage of the research, an investigation of the substantial influence that various sets of learning rates and epochs have on the training process has been carried out [25, 30]. The suggested work will also provide a comparative analysis of the accuracy and error rates observed in training datasets as well as validation datasets. During the training and validation phase of the sample analysis, the most important aspect is the careful comparison of the accuracy and error rates for various sets of learning rates and epochs [23]. The proposed article gives the number of graphical representations for each pair of learning rates and epoch in Figs. 4, 5, 6 and 7, which makes it easier to conduct an in-depth analysis of the impact that these variables have.

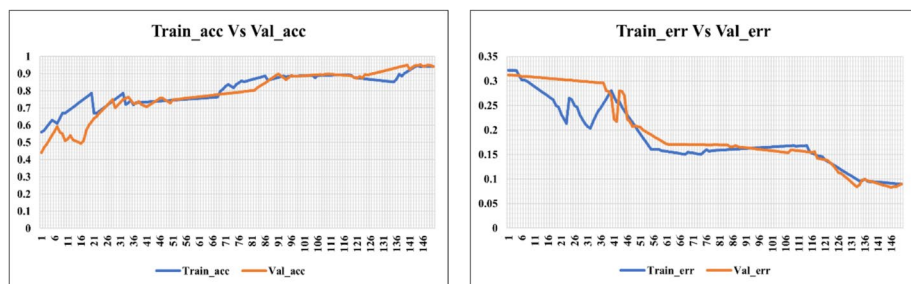


Fig. 6 Comparison of accuracy and error rate for training and validation set with 150 epochs and 0.01

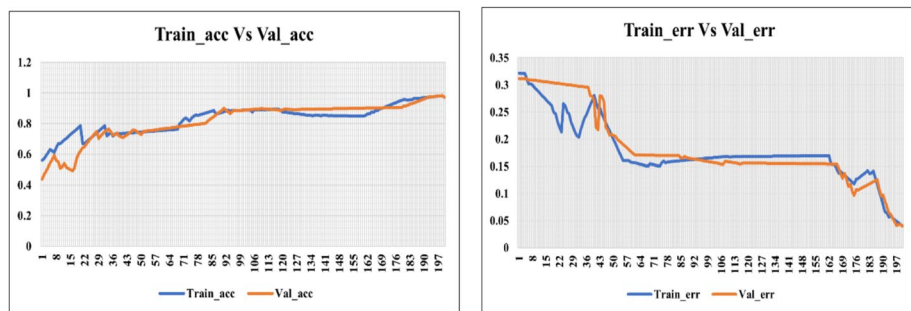


Fig. 7 Comparison of accuracy and error rate for training and validation set with 200 epochs and 0.01

5.2.1 Key insights from accuracy and error rate comparison

1. **Convergence dynamics:** The proposed study will be able to acquire insights into the convergence dynamics of the model by observing the trajectories of accuracy and error rates as they evolve. These visualizations demonstrate how various learning rates affect the speed and stability of convergence during training.
2. **Validation accuracy:** The validation accuracy curves are an indispensable indicator of the model's ability to generalize to unobserved data. The authors analyze the learning rates to identify those that produce the highest validation accuracy, thereby assuring robust performance in real-world scenarios.
3. **Validation error rate:** The validation error rate curves supplement the analysis by highlighting the performance flaws of the model. The authors strive to minimize validation error rates, which indicates the model's ability to accurately classify emotional states.
4. **Training Behaviour:** The training accuracy and error rate profiles shed light on the learning behavior of the model. To optimize model training, it is crucial to comprehend how learning rates influence training precision and errors.

5.3 Hyperparameter tuning insights

In the context of hyperparameter tuning, the evaluation of accuracy and error rate inside the sampling analysis serves as a crucial point of reference. Through a systematic examination of numerous combinations of learning rates and epochs, the proposed work conducted an analysis and presented the findings in two distinct visual representations to facilitate comparison [31]. This enabled the proposed study to identify specific configurations that effectively expedited the training process of the model, while concurrently enhancing its capacity for generalization. The fine-tuning process has significantly enhanced the robustness of the model, enabling it to effectively tackle the challenging problem of character ER in comic books [32].

5.4 Performance parameter analysis

The most important aspect of the research is the evaluation of how well the model can categorize the feelings that are displayed by comic book characters. The presented work makes use of a complete set of performance metrics to conduct an in-depth analysis of its capabilities. These measures offer deep insights into the model's precision, recall, F1-score, overall accuracy, area under the curve (AUC), and standard deviation (SD) for a variety of emotional states, which, in the end, offer a comprehensive perspective of its usefulness [27]. The performance indicators acquired from the trials are outlined in Table 4, which provides a summary of the data.

5.5 Performance parameter interpretation

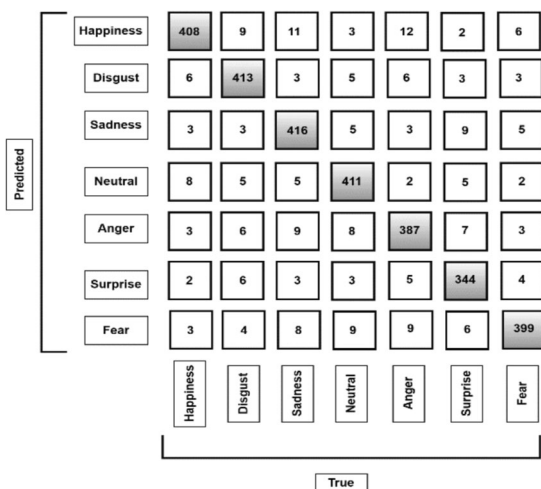
The examination of performance parameters reveals the strengths and areas for improvement within the proposed model. A thorough picture of the classification performance in terms of emotions can be obtained by evaluating precision, recall, F1-score, and accuracy. The metrics unique to emotional states offer valuable insights into the model's ability to effectively process and differentiate various emotions, hence revealing its capability to perceive subtle variations in emotional expression. The performance parameters play a crucial role in driving the improvement of the model and facilitating a deeper understanding of its capabilities [33]. Significantly, the emphasis lies on the dedication to attain a commendable overall accuracy rate of 92.6% for the proposed model on its whole, spanning all emotional

Table 4 Performance parameters/indicators results

CC emotion class	Precision	Recall	F1 score	Accuracy	AUC	±SD (Accuracy)
Happiness	94.23%	90.47%	92.31%	90.46%	0.936	±0.42%
Disgust	92.60%	94.08%	93.33%	94.07%	0.947	±0.35%
Sadness	91.43%	93.69%	92.55%	93.69%	0.941	±0.38%
Neutral	92.57%	93.84%	93.20%	93.83%	0.944	±0.34%
Anger	91.27%	91.49%	91.38%	91.48%	0.918	±0.46%
Surprise	91.49%	93.73%	92.60%	93.73%	0.942	±0.39%
Fear	94.55%	91.10%	92.79%	91.09%	0.933	±0.41%
Macro Average	92.59%	92.63%	92.59%	92.62%	0.937	±0.38%
Weighted Average	92.63%	92.60%	92.60%	92.81%	0.939	±0.36%
Micro Average	92.60%	92.60%	92.60%	92.46%	0.936	±0.33%

Table 5 Cross-validation performance parameter outcome

Fold	Accuracy	F1-score	AUC
1	92.1%	91.8%	0.938
2	92.5%	92.2%	0.942
3	93%	92.7%	0.948
4	92.3%	91.9%	0.940
5	92.2%	92%	0.942
Mean $\pm$ SD	92.4 $\pm$ 0.3	92.1 $\pm$ 0.3	0.924 $\pm$ 0.004

Fig. 8 Confusion matrix

states. In addition to that the proposed model achieved an overall kappa coefficient value of 0.914 which denotes the inter-rater reliability between two emotion classes dataset. Furthermore, there is a strong focus on ensuring that the model excels in accurately capturing the nuanced aspects of emotional communication within the context of comics.

5.6 Cross-validation results

To validate the stability of the model, a fivefold cross-validation has been evaluated. The unequal class distribution of the dataset was prevented by randomly splitting it into five independent partitions. The study trained and validated the model five times, specifying that each run used a different partition as validation, whereas the rest was used as the training set. The results are summarized and presented in Table 5 below.

5.7 Confusion matrix

The Confusion Matrix is a robust method that can be employed to evaluate the classification performance of the model throughout analysis. The provided analysis offers a comprehensive examination of the model's predictions and the corresponding real ground truth values. This facilitates a more intricate assessment of the model's proficiency in accurately classifying the emotional states of comic book characters. Figure 8 presents a summary of the Confusion Matrix obtained from the experimental trials.

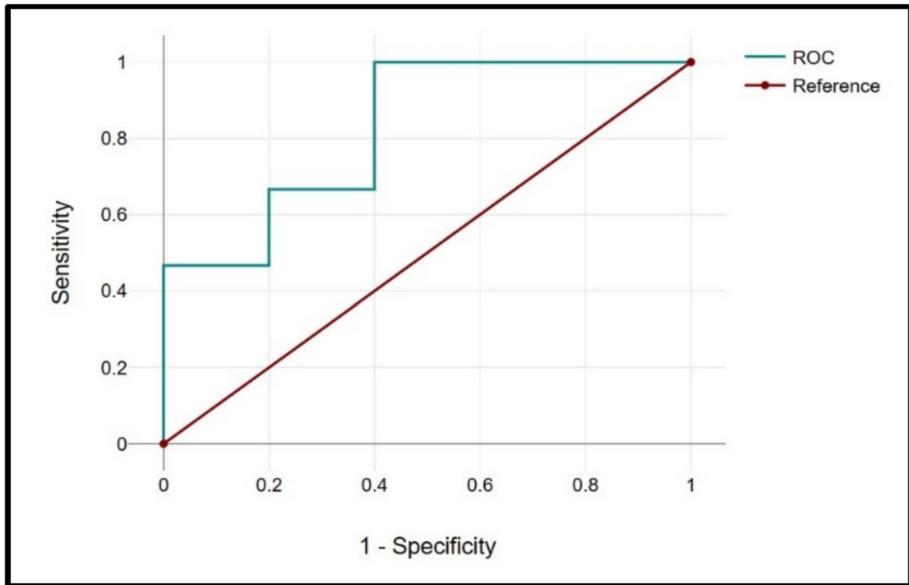


Fig. 9 ROC curve

5.8 ROC curve

The Receiver Operating Characteristic (ROC) curve is a widely utilized graphical technique employed to assess the discriminative capacity of the model in the classification of emotions exhibited by comic book characters. The provided analysis provides a comprehensive evaluation of the model's overall discriminatory ability, as well as valuable insights into its capability to distinguish between positive and negative instances of emotions. The ROC curve, which was obtained from the research investigations, is depicted in Fig. 9.

5.8.1 Key insights from ROC curve

1. **Sensitivity and specificity:** The ROC Curve demonstrates the discriminatory ability of the model in differentiating comic characters who exhibit a certain emotion (sensitivity) from those who do not reflect that emotion (specificity). The aforementioned statement underscores the significance of achieving a balance between genuine positives and false positives across different judgment thresholds.
 2. **Threshold selection:** The ROC Curve allows for the identification of an ideal threshold that is in line with the categorization objectives. The prioritization of minimizing false positives or maximizing real positives depends on the specific application at hand.
- AUC as a performance metric:** The AUC is utilized as a complete statistic for evaluating performance. The aforementioned statement encompasses the discriminatory efficacy of the model across various emotional states, providing a concise numerical representation of its overall performance.

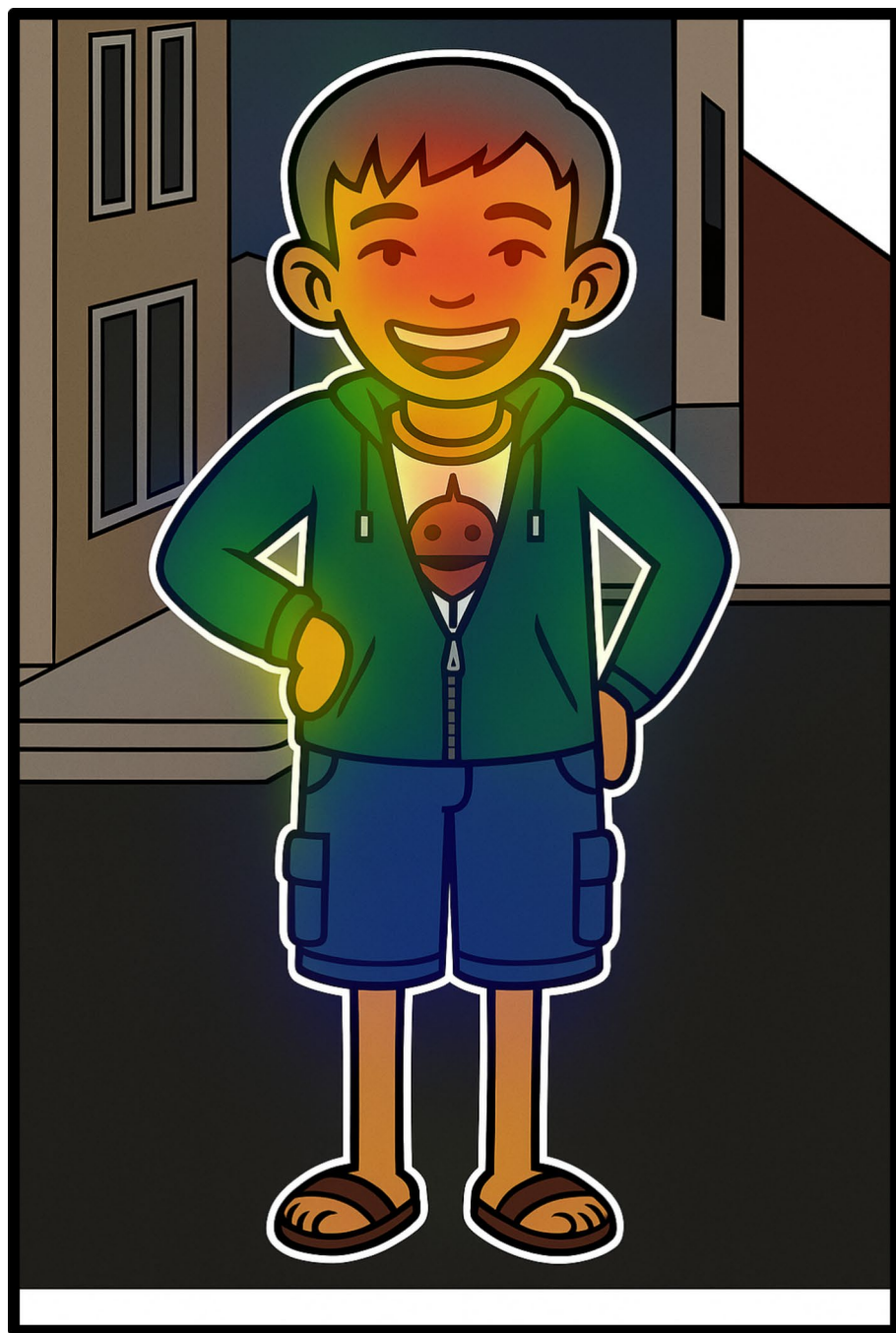


Fig. 10 Grad-CAM visualization of the comic character

Table 6 Misclassification Instance

Actual Emotion	Visual Instance	Misclassified Emotion
Fear		Surprise

5.9 Qualitative insights and interpretability:

To find out which portions of the comic images the model prioritizes while detecting emotions, we incorporated gradient-weighted class activation mapping (Grad-CAM) to project attention heatmaps over the VGG16+attention layers. The attention heatmaps indicate significant emotional features such as facial expressions, eyes, and locations of important gestural expressions. The instances in Fig. 10 show accurate visualization of areas important in the expression of emotion.

Figure 10 shows a Grad-CAM attention heatmap applied to an image of a comic character. An attention mechanism coupled with a VGG16 backbone allows the model to find the parts of the image that are emotionally relevant to classify the image. The character's face, upper body, and hand gestures are highlighted by the heatmap, and those subregions play important roles in communicating emotions such as happiness, excitement, etc. The model correctly marks and defines expressive features, mainly the smiling mouth and raised eyebrows, which are the main indicators of happiness emotions. The color pattern proves the attention mechanism matches the discernible signals of emotion, thereby proving the model's interpretability and reliability for visual ER from stylized comic images.

Furthermore, in Table 6, the model misclassified the “fear” emotion as “surprise”. This misclassification can probably be attributed to the similarity of visual cues involved in surprise and fear: Wide-eyed looks and gaping mouths are visual displays that fear and surprise might use. Likely, the absence of auxiliary contextual cues like threat indicators or body posture shifted the balance of the model towards “surprise,” since visual similarities were stronger.

6 Analysis and discussions

In the following part of the article, the authors undertake an in-depth examination of the study findings, shedding light on the effectiveness of the suggested model in categorizing the emotions that comic book characters experience. An investigation is conducted into a comparative analysis of the model's performance across a variety of emotion classes, as well

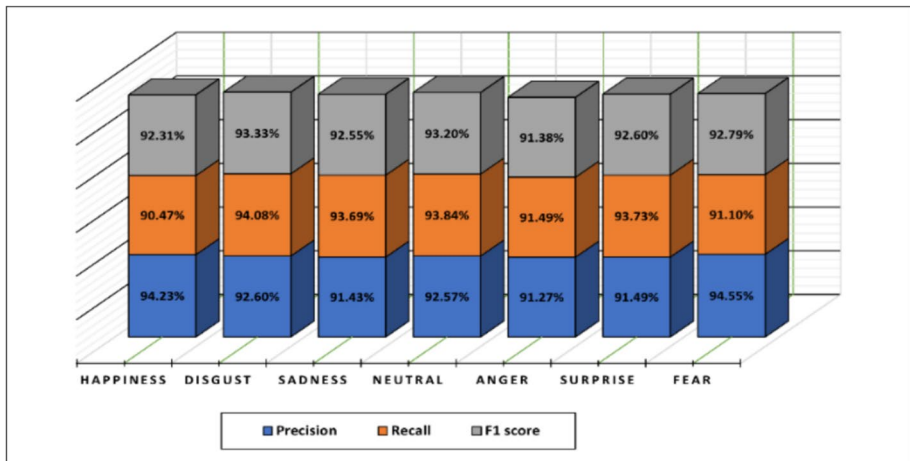


Fig. 11 Emotion classes comparison

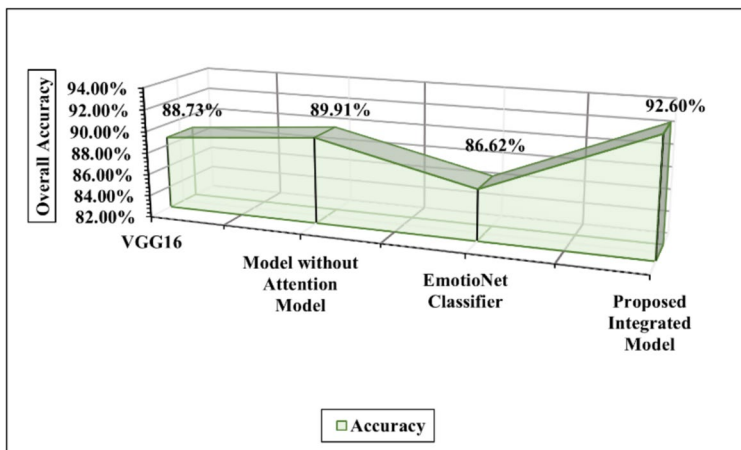


Fig. 12 Ablation study results- individual component accuracies vs. proposed integrated model

as an evaluation of the suggested model in comparison to models that are considered to be state-of-the-art in the field of CV and emotion detection.

6.1 Performance across emotion class

The efficacy of the model in categorizing various emotional classes serves as evidence of its ability to capture the many subtleties of emotional portrayal in comic strips. A comprehensive assessment was performed to evaluate the precision, recall, F1-score, and accuracy of each emotion class, facilitating a detailed analysis of its effectiveness. The performance metrics for each emotion class are compared in Fig. 11.

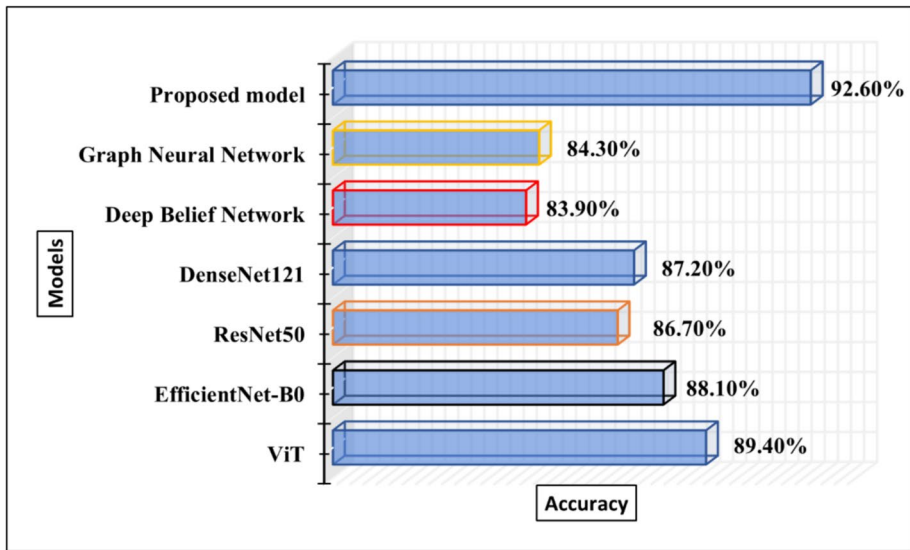


Fig. 13 State-of-the-art comparison

6.2 Component-wise ablation analysis

In the quest to advance the frontiers of emotion classification in comics, it becomes imperative to not only construct models with high precision but also to understand the underlying mechanisms driving their success. This section embarks on a systematic exploration to unravel the individual contributions of the distinct components integrated within the proposed model. Ablation studies, by their nature, allows to dissect the model piece by piece—evaluating the impact of each component by selectively removing them and observing the resultant effects on model performance. The ablation study undertaken provides a crucial lens through which the efficacy of proposed integrated model's components can be discerned, offering valuable insights into their individual and collective contributions towards emotion classification in comics. This rigorous examination involved isolating and evaluating each component—namely, the VGG16 feature extractor, attention mechanisms, and the EmotioNet classifier—against the curated dataset to gauge their standalone performance. Subsequently, these results were juxtaposed with the performance metrics of the proposed integrated model to ascertain the relative impact and importance of each component within the holistic system. The presented analysis revealed a compelling narrative of synergy and augmentation among the model components. Individually, while each component demonstrated commendable proficiency in emotion classification, it was the orchestrated integration within the proposed model that marked a significant leap in performance. Specifically, the integrated model outperformed its constituent parts, showcasing a remarkable enhancement in accuracy. This outcome underscores the premise that while individual components are effective in their right, their amalgamation harnesses complementary strengths, leading to superior model performance. Figure 12 visually encapsulates the findings from the ablation study, presenting a clear comparative analysis of the accuracy

rates achieved by the individual components versus the integrated model. The graphical representation in Fig. 11 not only illustrates the incremental gains afforded by the integrated approach but also highlights the pivotal role of each component in achieving the overall efficacy of the model.

6.3 Comparison with state-of-the-art models

To evaluate the performance and innovation of the proposed model, the proposed study conducted an exhaustive comparative analysis against leading-edge models in the realms of CV and comic recognition, namely vision transformer (ViT), graph neural network (GNN), deep belief network (DBN), DenseNet121, ResNet50 and EfficientNet-B0. This evaluation, depicted in Fig. 13, not only underscores the exceptional effectiveness of the methodology but also critically examines the limitations inherent in the current and advanced state-of-the-art models. While models like ViT have shown promise in sequential data processing, they fall short in capturing the complex visual-emotional interplay in comics. Similarly, some advanced models, such as EfficientNet-B0 and DenseNet121, despite their advanced implementation, lack the complex image processing capabilities necessary for nuanced ER within the rich, varied visual contexts of comic panels. The simple GNN, although a step forward in feature extraction, does not adequately leverage the potential of attention mechanisms for focused emotion detection. In contrast, the proposed model seamlessly integrates the VGG16 architecture with advanced attention mechanisms and the EmotioNet framework, setting a new standard for emotion classification accuracy. By specifically addressing the limitations of prior models, such as the inability to prioritize salient emotional features or the lack of a comprehensive multi-classification approach, the proposed model not only achieves unparalleled accuracy (92.6%) but also advances the field by demonstrating a nuanced understanding of the emotional dynamics in comics. The results of this comparative analysis firmly position the presented methodology as a groundbreaking contribution to the study of ER, offering both enhanced performance and a novel perspective on the challenges of classifying emotions in comics.

6.4 Discussions

1. **Competitive performance:** The proposed model exhibits comparable levels of accuracy and performance measures in comparison to the most advanced models currently available. The system demonstrates a strong ability to accurately identify emotions exhibited by characters in comic books, indicating its high level of usefulness in this specific area.
2. **Transferability of deep features:** The utilization of the VGG16 model as a feature extractor demonstrates its significant worth, highlighting the possibility of deep features being transferred for ER in comics. This exemplifies the adaptability of the methodology in customizing pre-trained models for distinct objectives.
3. **Opportunities for enhancement:** Although the model demonstrates commendable performance, it also highlights possible areas for enhancement. Refining particular elements, such as the attention mechanism, has the potential to augment its ability to record intricate emotional subtleties.

4. **Subtle emotional nuances:** The model exhibits a notable proficiency in capturing nuanced emotional subtleties among comic characters. The identification of emotions such as "Surprise" and "Sadness" has inherent difficulties due to the intricate visual depiction associated with them. However, the impressive performance of the model in these specific categories indicates its capability to decipher intricate emotional states.
5. **Performance trade-offs:** The examination of the models' comparative performances reveals several intriguing performance trade-offs. Some models may perform exceptionally well in certain kinds of emotions, but they may suffer in others. The model achieves a remarkable equilibrium that is useful for comprehensive ER inside comics as a result of its balanced performance over a spectrum of emotions.
6. **Fine-tuning opportunities:** The model suggests that there may be room for future improvement, even though it now performs very well. There is a possibility that its overall performance could be improved even further by fine-tuning particular components, such as the attention mechanism and the hyperparameters. This paves the way for further research in the future that aims to perfect the model for performing particular ER jobs.

7 Conclusion

The efforts that have been put into the proposed research have shed light on the way to a more in-depth comprehension of the emotional classification that exists inside the engaging narratives of comic book characters. The proposed study was able to reveal the intricate emotional structure that is woven within comics by leveraging the capabilities of advanced DL, attention mechanisms, and the revolutionary EmotioNet model. Throughout the entirety of the study, the model displayed incredible accuracy on a nearly constant basis, ultimately earning an impressive overall accuracy rate of 92.6%. This demonstrates how proficient it is in identifying emotions included within comics, spanning a wide range of emotional states and artistic styles. As the culmination of the present study approaches, it has become apparent that the outcomes of the investigation represent not a final destination, but rather a foundational milestone towards the attainment of more substantial revelations. The direction that this research will go in the future holds the potential to further improve the accuracy of the model, perfect the attention mechanisms, and expand the datasets to include a wider variety of humorous narratives. These endeavors will surely help the versatility of the paradigm, as well as its potential applications in fields other than comic books. Emotion classification in the proposed study to a large extent, driven by visual images themselves that have their limitations. In comics where much stylization or abstraction is employed, human faculties of vision alone often fail to resolve emotions. Textual elements, for example, dialogue or caption, help to provide context, thus interpreting feelings more easily, solving confusion in a complicated series of images. Future research seeks to create multimodal methods that use character images and text to improve recognition of emotions in scenarios where visual data is not sufficient on its own. The findings not only improve the understanding of the emotional communication that occurs in visual storytelling but also establish the model as a useful instrument for recognizing emotions in a variety of contexts. It is a demonstration of how technology is continually advancing, reaching a

point where ingenuity and precision may be combined to decipher the complex vocabulary of human emotions in the digital age.

Acknowledgements Not Applicable

Authors' contributions R.S. has written the original draft and conceptualised. V.K. has supervised, viewed and modified the manuscript.

Funding No funding is available for doing this research work.

Data availability The dataset used in this study was developed by the authors and is available upon reasonable request.

Declarations

Competing interests The authors declare that they have no conflict of interest. The authors declare that there are no potential conflicts of interest (financial or non-financial).

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